Software Tools and Technologies Laboratory

Latex Lab Activity - 3: Beamer

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15th September, 2025



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2. Why Beamer?
3. Features
4. Highlighting Texts
5. Mathematical Concepts
6. Linear Search Algorithm
7. Images
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Introduction to LaTeX Beamer

- LaTeX Beamer is a powerful class for creating presentations.
- It allows:
  - Professional-quality slides
  - Mathematical equations
  - Figures, tables, and references
  - Animations and overlays
- Advantages:
  - Consistent formatting
  - Easy handling of citations and references
  - Cross-platform PDF output



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- Free and open-source — no licensing issues



Comparison with Other Tools

Feature	Beamer	PowerPoint
Equations	Excellent	Limited
Citations		
Portability		
Cost		
Customization		



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Customization	High (code-based)	Moderate (GUI-based)



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Key Features of LaTeX Beamer

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- Predefined professional themes for elegant slides

Content Capabilities

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- Integration with BibTeX for automatic citations
- Efficient workflow for large, structured presentations

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Blocks and Highlights Example

Some important text can be highlighted to attract attention.

Remark

This is a normal block. It wraps the text in a box with the same style as the presentation. The text after the block title is the content.

Important Theorem

This is an alert block. It contrasts with the regular presentation style, useful to emphasize key results or theorems.

Examples

This is an example block. It is similar to a normal block, but has a different style (less contrasting than alertblock). Perfect for examples.



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Definition

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A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.



Theorem & Lemma

Theorem

Fundamental Theorem of Arithmetic: *Every integer greater than 1 is either a prime or can be written as a unique product of prime numbers.*

Lemma

If a prime p divides a product ab, then p divides a or p divides b.



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Linear Search Algorithm

Input: Array A of size n, key key to search
Output: Index of key in A, or -1 if not found
for $i \leftarrow 0$ **to** $n - 1$ **do**
    if $A[i] = key$ **then**
        return i;
    end
end
return -1;

Algorithm 1: Linear Search



Corollary & Proof

Corollary

The factorization of any integer into primes is unique up to the order of the factors.

Proof.

Suppose n has two different prime factorizations. By the lemma, every prime in one factorization divides one of the primes in the other factorization. This leads to a contradiction, proving uniqueness. □



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Including an Image in a Block

Example Image



Figure: This is a sample image included in a Beamer slide.

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Key Equations in Signal Processing

Equation 1: Fourier Transform

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\, dt$$

Equation 2: Convolution Integral

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau)\, d\tau$$

Equation 3: Transfer Function of LTI System

$$H(f) = \frac{Y(f)}{X(f)} = \int_{-\infty}^{\infty} h(t) e^{-j2\pi ft}\, dt$$



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- Highly customizable themes, colors, and backgrounds make slides visually appealing.



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- It supports complex mathematical equations, figures, tables, and references.
- Overlays and incremental reveals help guide the audience's focus.
- Highly customizable themes, colors, and backgrounds make slides visually appealing.
- Ideal for academic and technical presentations due to its precision and portability.



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References I

This presentation is based on the following references:

- [4] for basic LaTeX concepts.
- [2] for fundamental TeX understanding.
- [1] for advanced LaTeX usage.
- [5] for best practices in document preparation.
- [3] for graphics and presentation guidelines in LaTeX.


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Thank You!

Any Questions?



